

Assignment - 1.3

Day 2 : RAM and Storage Theory : Types of RAM (DDR3, DDR4, DDR5) and storage (HDD vs SSD, NVMe). Practical :- Remove and reinstall RAM and storage drives.

Task – 1

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800–8800+MT/s
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1V
Capacity	Up to 16 GB	Up to 64 GB	48-128 GB+
Typical Use Cases	Older desktops, laptops, legacy servers	Mainstream PCs, gaming systems, workstations (2015–2023 era)	Modern gaming PCs, high-performance workstations, AI/content creation, and newest desktop/server platforms

In Summary :-

DDR3 :- Maintaining or upgrading older systems & Lowest cost for legacy hardware

DDR4 :- Most existing PCs and budget-to-midrange builds & Excellent value and broad compatibility.

DDR5 :- New high-performance systems and Highest speed & efficiency

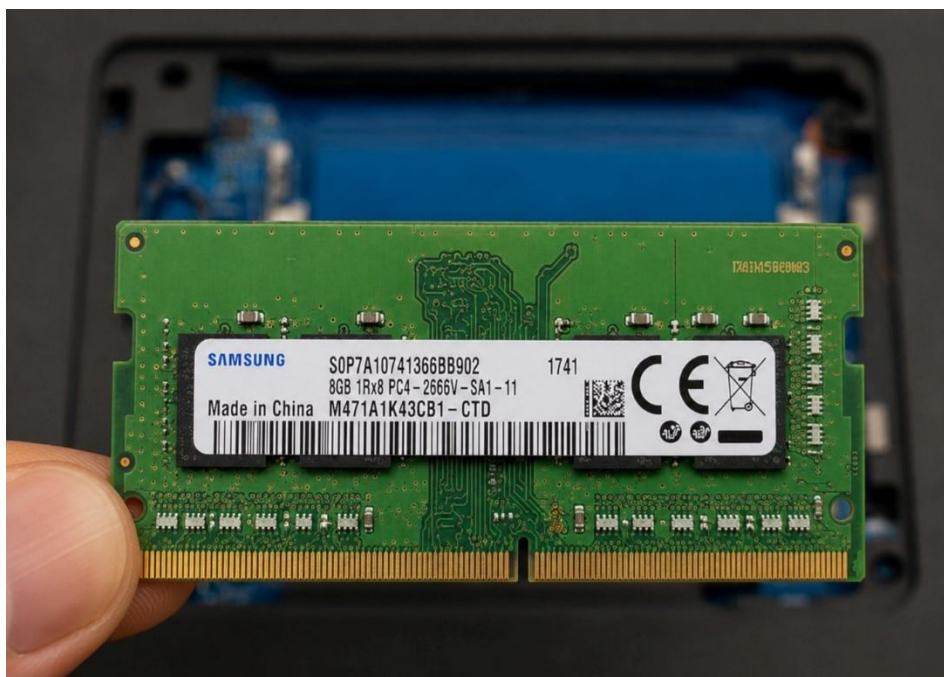
Task – 2

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Step 1 :- RAM Slots (Power Disconnected)



Step 2 :- Installed RAM Module



Step 3 :- RAM Module Reinstalled



Step 4 :- System Boots Successfully :-

Device specifications

Device name	LAPTOP-9B0K9Q6E
Processor	Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz 1.80 GHz
Installed RAM	8.00 GB (7.87 GB usable)
Device ID	D1C5E3D0-63D0-4B0F-8E1E-2A6B1A9D2E3F
Product ID	00327-35862-23330-AAOEM
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Windows specifications

Edition	Windows 10 Home Single Language
Version	22H2
Installed on	14-05-2024
OS build	19045.4529
Experience	Windows Feature Experience Pack 1000.19056.1000.0

Task – 3

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Features	HDD	SSD	NVMe SSD
Speed	Slow	Fast	Very Fast
Main Difference	Uses spinning magnetic disks; cheaper and offers large storage	Uses flash memory with no moving parts; faster and more reliable than HDD	Uses PCIe interface, making it much faster than a normal SSD
Best Daily Use Case	Google Drive/Photos backup – useful for storing large amounts of files and photos	Windows + Microsoft Office – apps open and run faster	Gaming (e.g., PUBG/Free Fire) – gives faster game loading and level loading

Three key differences:

1. **Speed:** HDD < SSD < NVMe SSD
2. **Technology:** HDD uses spinning disks, while SSD and NVMe use flash memory.
3. **Connection & performance:** NVMe uses PCIe and provides the highest performance, while regular SSDs commonly use SATA.

Task – 4

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

1. Locate the Storage Drive



2. Drive Removed



3. Connector Type (SATA)



4. Drive Reinstalled



5. System Booted Successfully



6. Drive Recognized in Windows (Disk Management)

